

University of Vavuniya
Faculty of Applied Science
Department of Physical Science

IT 3223 Advanced Database Management Systems (Practical)

In-Course Assessment – 01

Time: 40 min

Date: 30.03.2026

Assume you are tasked with developing a simple application for a library to manage its book lending operations. The library database is structured to store and manage information about members who borrow books (*Members: MemberID, Name, Address, City*), the books available in the library (*Books: BookID, Title, AuthorID, Price, Publisher*), and the authors who wrote those books (*Authors: AuthorID, Name, Country*). It also includes a Borrowing_Books table to record borrowing and returning activities (*Borrowing_Books: MemberID, BookID, BarrowedDate, ReturnedDate*). Furthermore, the database defines the following relationships: each book is associated with a single author, each member can borrow multiple books, and the Borrowing_Books table serves as a link between members and books.

1. Create a user with your registration number as the user name and with a password. [05%]
2. Grant all privileges to the created user. [10%]
3. Login using the created user and create the library database by including all the relations mentioned in the description by applying all the primary key and foreign key constraints. [15%]
4. Answer the following questions with respect to the view concept in MySQL: [70%]
 - i. Create a view to display all information about the books borrowed by each of the members.
 - ii. Create a view to get the author name and publisher for each of the book title.
 - iii. Create a view to list down the currently borrowed book names.
 - iv. Create a view to find the number of books borrowed by each of the members.
 - v. Create a view to get the expensive books (books with a price more than the average price).
 - vi. Create a view to find the books written by each of the authors.
 - vii. Create a view to find the total price of the books borrowed by each of the members.
 - viii. Create a view to find the book name and author name for the book with the minimum price.
 - ix. Create a view to display the borrowed date and returned date for each of the books.
 - x. Create a view to get the details of authors from the UK and all the books written by them.

- xi. Insert a new record into the Borrowing_Books table with the following values: 'M05', 'B10', '2024-03-03', '2024-03-13'. Describe your observation when selecting all the records from the view created in Part (a).
- xii. List down all the views created by you for the library database.
- xiii. Delete the view created for Question 9 and list down all the views created by you again.

Note: The soft copy of the data file is provided with the necessary information.

Members			
MemberID	Name	Address	City
M01	John Silva	Colombo 1	Colombo
M02	Anu Perera	Kandy Rd	Kandy
M03	Ravi Kumar	Jaffna Town	Jaffna
M04	Nimal Fernando	Galle Fort	Galle
M05	Sita Devi	Matara Rd	Matara
M06	Kamal Das	Negombo	Negombo)
M07	Sunil Raj	Kurunegala	Kurunegala
M08	Priya Nair	Trincomalee	Trinco
M09	Rohan Peris	Kalutara	Kalutara
M10	Mala Silva	Badulla	Badulla

Books				
BookID	Name	AuthorID	Price	Publisher
B01	Harry Potter	A01	1500	Bloomsbury
B02	1984	A02	1200	Penguin
B03	Animal Farm	A02	1000	Penguin
B04	Tom Sawyer	A03	1300	Classic Pub
B05	The Guide	A04	1100	Indian Press
B06	The Alchemist	A05	1400	HarperOne
B07	Da Vinci Code	A06	1800	Doubleday
B08	Pride and Prejudice	A07	1250	Modern Pub
B09	War and Peace	A08	2000	Russian House
B10	Murder on the Orient Express	A10	1600	Collins

Authors		
AuthorID	Name	Country
A01	J.K. Rowling	UK
A02	George Orwell	UK
A03	Mark Twain	USA
A04	R.K. Narayan	India
A05	Paulo Coelho	Brazil
A06	Dan Brown	USA
A07	Jane Austen	UK
A08	Leo Tolstoy	Russia
A09	Ernest Hemingway	USA
A10	Agatha Christie	UK

Borrowing_Books			
MemberID	BookID	BorrowedDate	ReturnedDate
M01	B01	2024-01-01	2024-01-10
M02	B02	2024-01-02	2024-01-12
M03	B03	2024-01-03	2024-01-13
M04	B04	2024-01-04	2024-01-14
M05	B05	2024-01-05	2024-01-15
M06	B06	2024-01-06	2024-01-16
M07	B07	2024-01-07	2024-01-17
M08	B08	2024-01-08	2024-01-18
M09	B09	2024-01-09	2024-01-19
M10	B10	2024-01-10	2024-01-20
M01	B02	2024-02-01	2024-02-10
M02	B03	2024-02-02	2024-02-12
M03	B04	2024-02-03	null)
M04	B05	2024-02-04	2024-02-14
M05	B06	2024-02-05	2024-02-15
M06	B07	2024-02-06	2024-02-16
M07	B08	2024-02-07	null
M08	B09	2024-02-08	2024-02-18
M09	B10	2024-02-09	2024-02-19
M10	B01	2024-02-10	2024-02-20